
Markov Decision Process

– II

CS5491: Artificial Intelligence

ZHICHAO LU

Content Credits: [Prof. Wei's CS4486 Course](#)
and [Prof. Boddeti's AI Course](#)



TODAY

Value Iteration

Policy Iteration

- Policy Evaluation
- Policy Extraction



EXAMPLE: GRID WORLD

A maze like problem

- The agent lives in a grid
- Walls block the agent's path

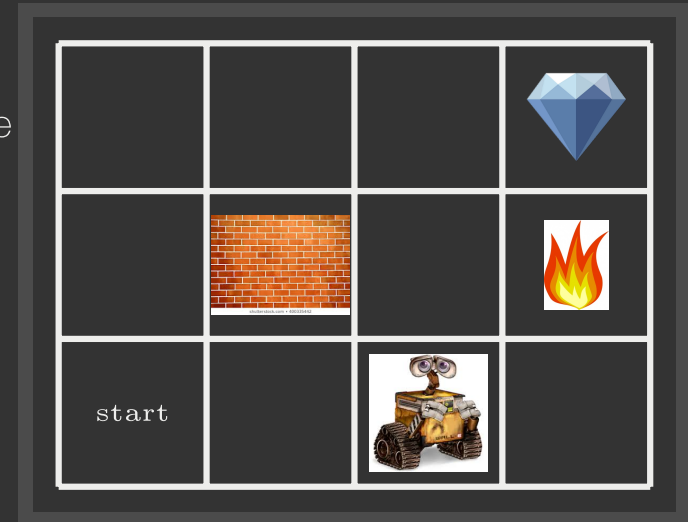
Noisy movement: actions do not always go as planned

- 80% of the time, the action North takes the agent North (if there is no wall there)
- 10% of the time, North takes the agent West; 10% East
- If there is a wall in the direction the agent would have been taken, the agent stays put

The agent receives rewards each time step

- Small "living" reward each step (can be negative)
- Big rewards come at the end (good or bad)

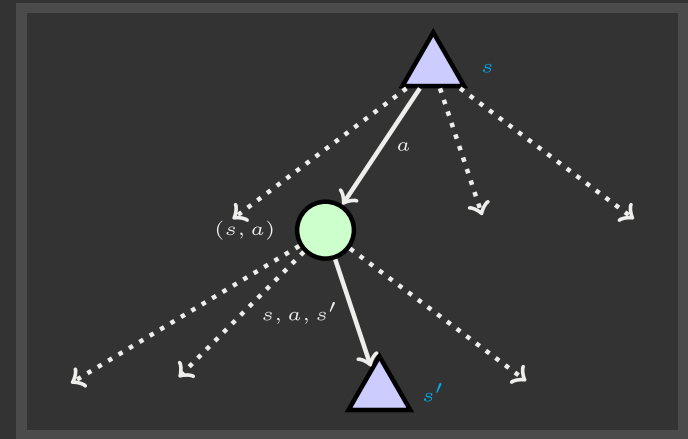
Goal: maximize sum of rewards



RECAP: DEFINING MDPs

Markov decision processes:

- Set of states $\mathcal{S}$
- Start state s_0
- Set of actions $\mathcal{A}$
- Transitions $P(s'|s, a)$ (or $T(s, a, s')$)
- Rewards $R(s, a, s')$ (and discount γ)



MDP quantities so far:

- Policy = Choice of action for each state
- Utility = sum of (discounted) rewards

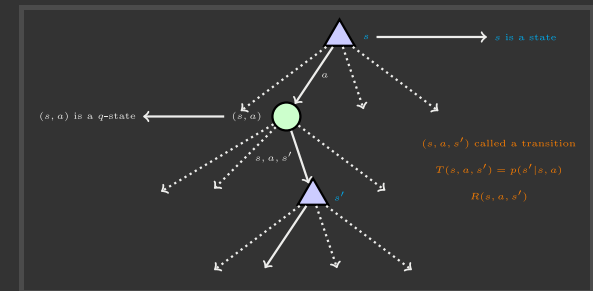


SOLVING MDPS



OPTIMAL QUANTITIES

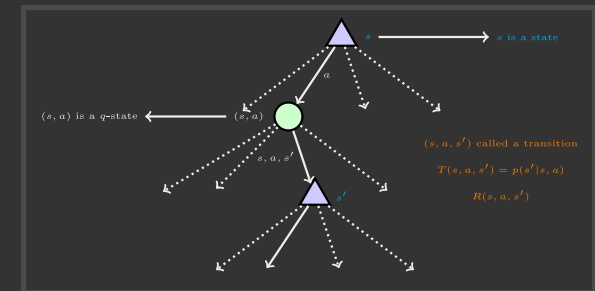
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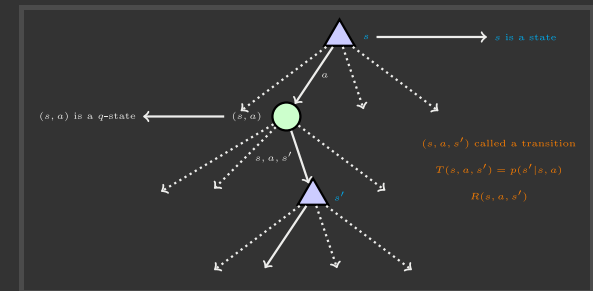
The value (utility) of a $\mathbf{q}$ -state $(\mathbf{s}, \mathbf{a})$: $Q^*(\mathbf{s}, \mathbf{a}) =$ expected utility starting out having taken action $\mathbf{a}$ from state $\mathbf{s}$ and (thereafter) acting optimally



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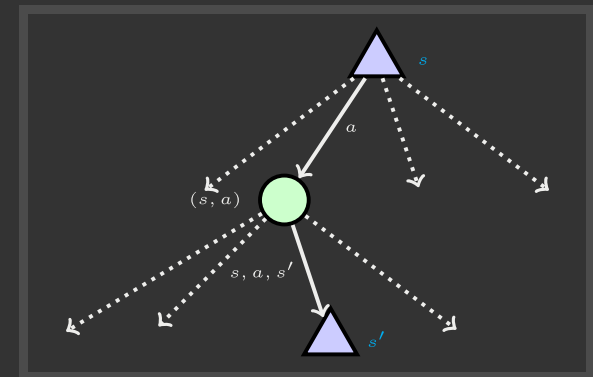


The optimal policy: $\pi^*(\mathbf{s}) =$ optimal action from state $\mathbf{s}$

VALUES OF STATES

Fundamental operation: compute the (expectimax) value of a state

- Expected utility under optimal action
- Average sum of (discounted) rewards
- This is just what expectimax computed

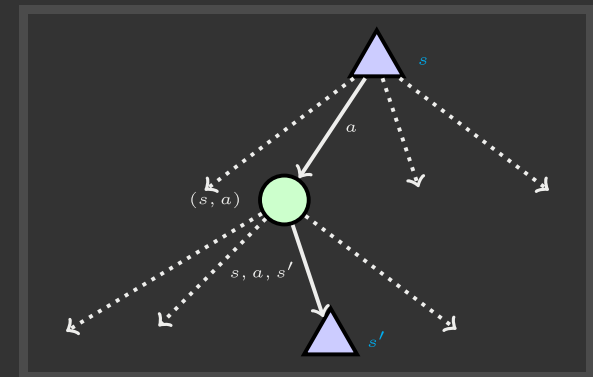


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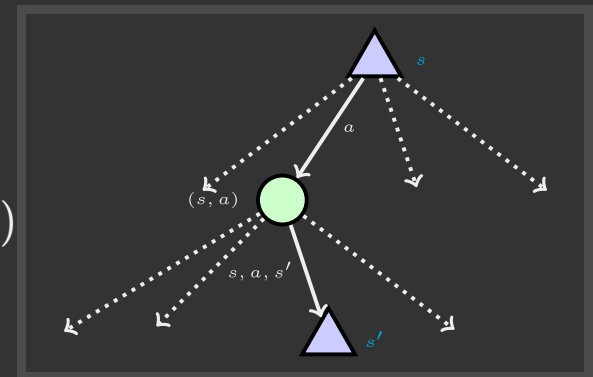
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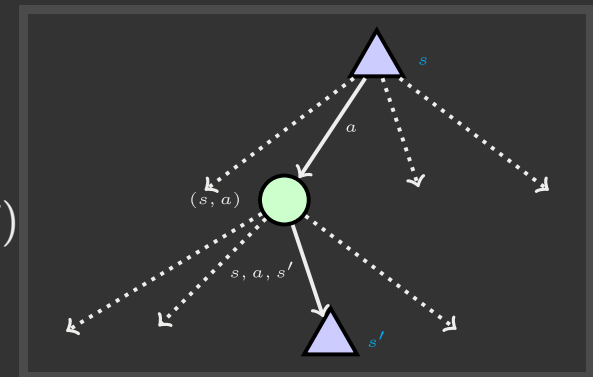
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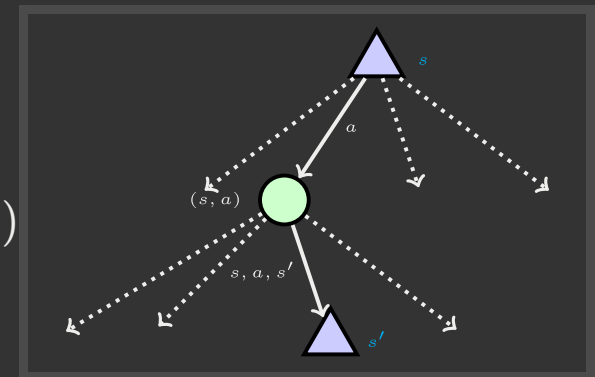
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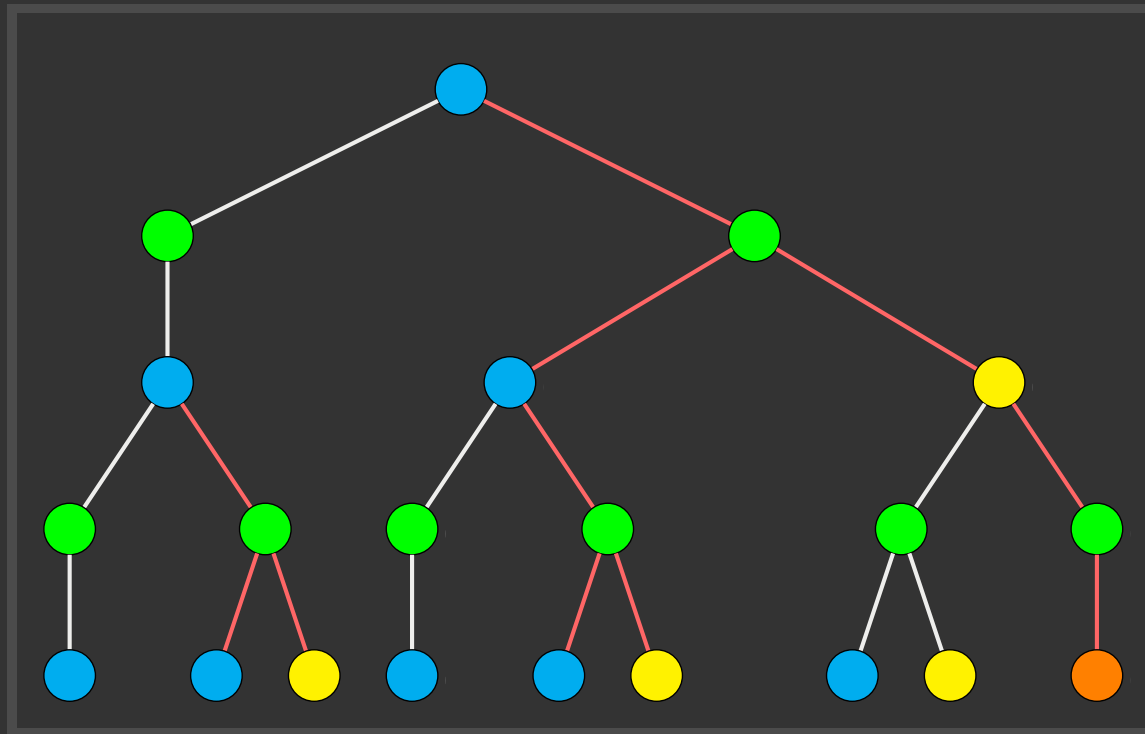
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Problem 2: Tree goes on forever

- Idea: Do a depth-limited computation, but with increasing depths until change is small
- Note: deep parts of the tree eventually do not matter if $\gamma < 1$



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Key idea: time-limited values



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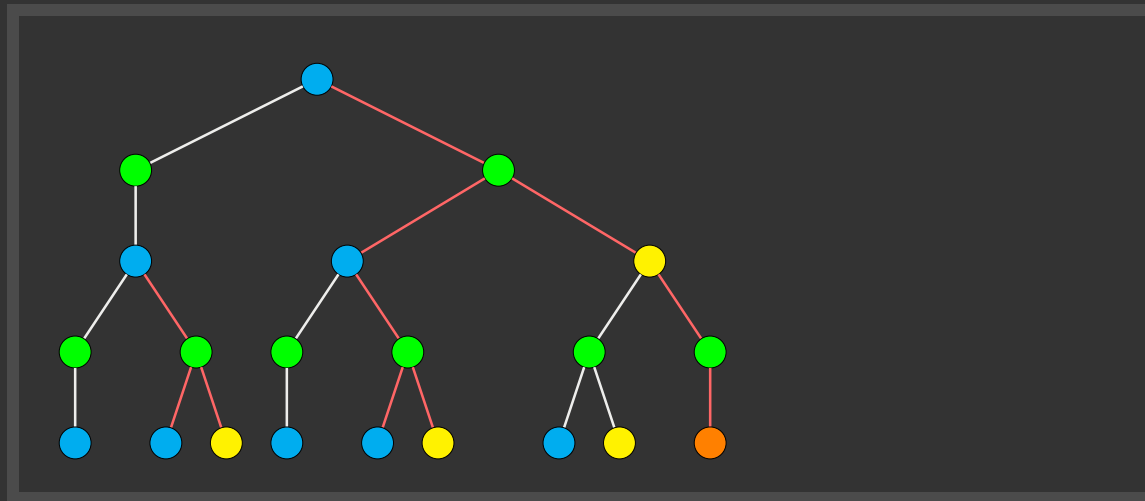


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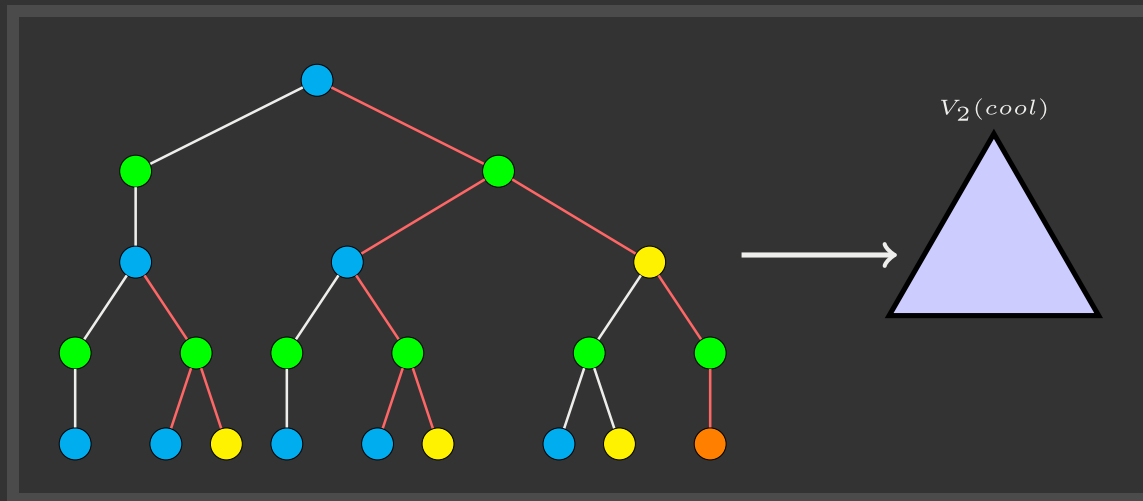


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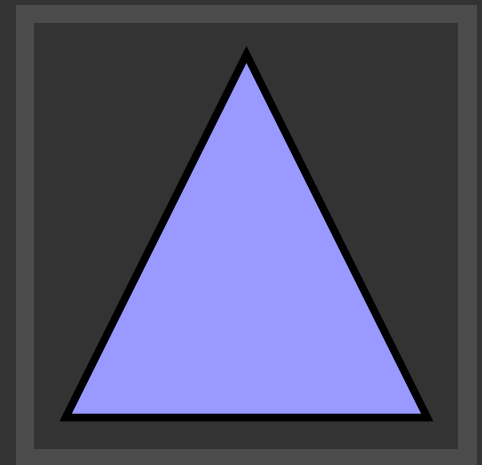
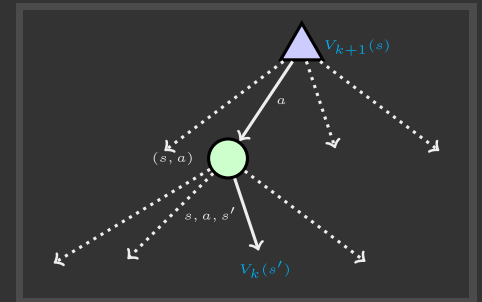


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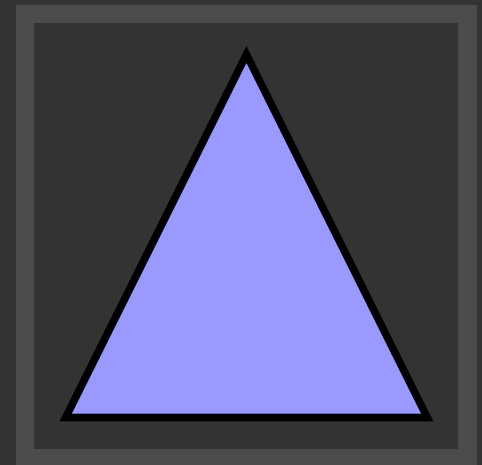
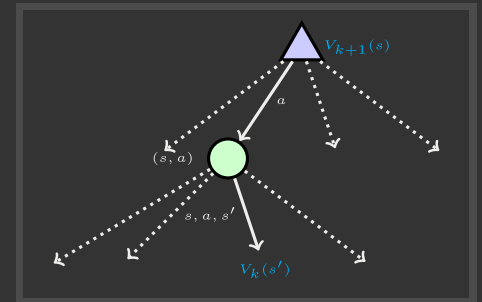


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Given vector of $V_k(s)$ values, do one ply of expectimax from each state:

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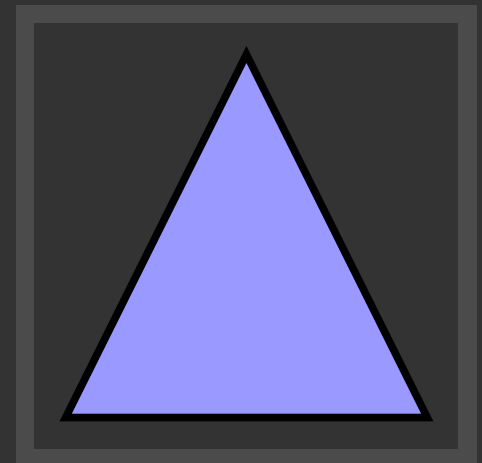
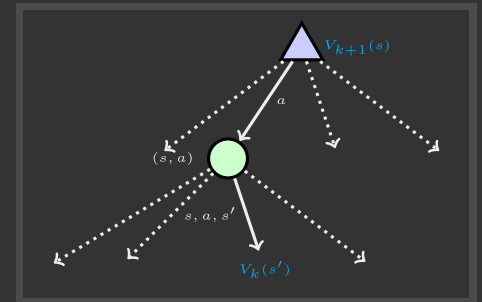
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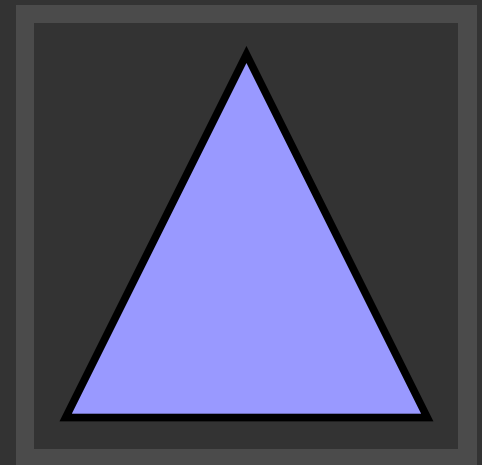
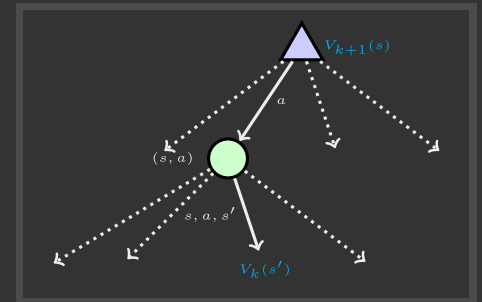
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Complexity of each iteration: $\mathcal{O}(S^2 A)$



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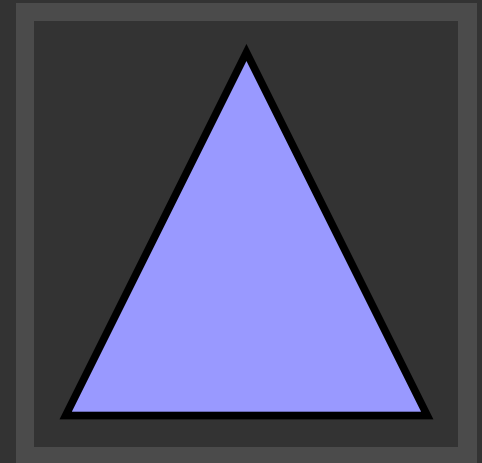
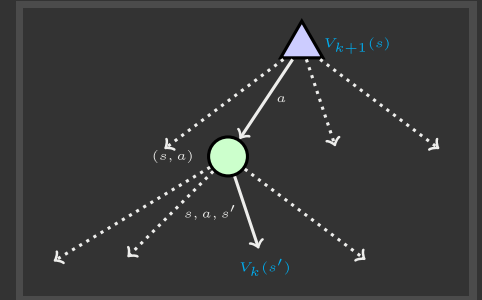
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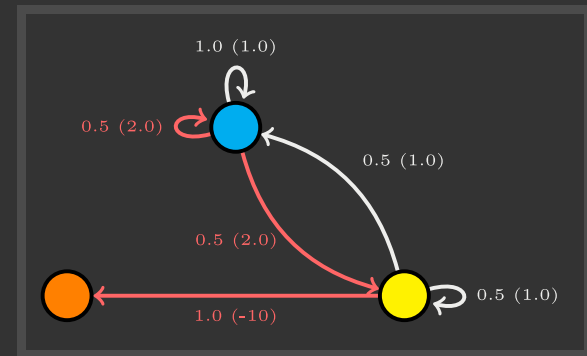
Theorem: will converge to unique optimal values

- Basic idea: approximations get refined towards optimal values
- Policy may converge long before values do



EXAMPLE VALUE ITERATION

	Cool	Warm	Overheated
V_2			
V_1			
V_0	0	0	0

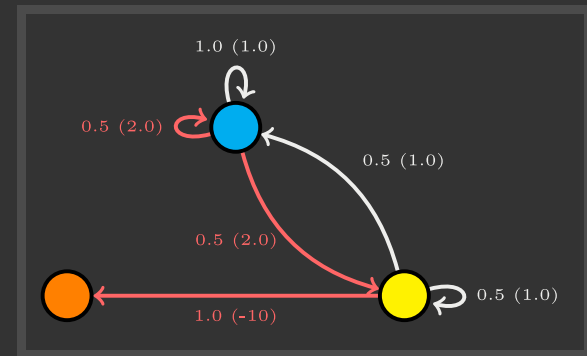


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Assume no discount

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V_1	2	1	0
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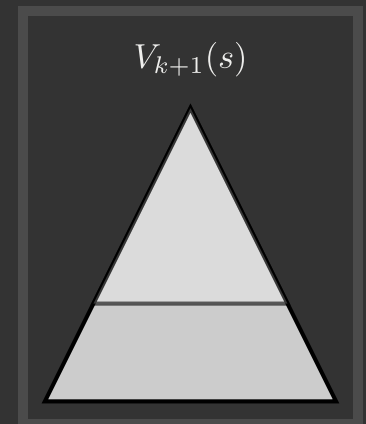
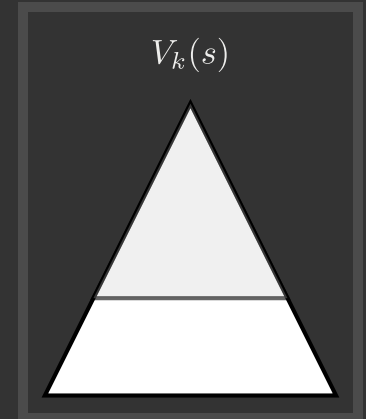
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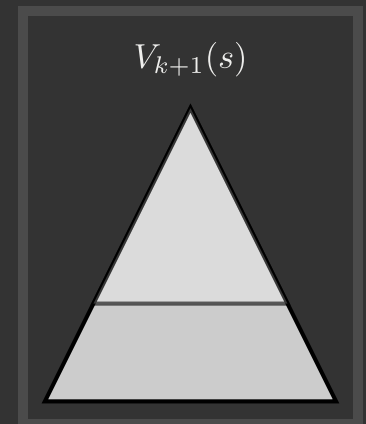
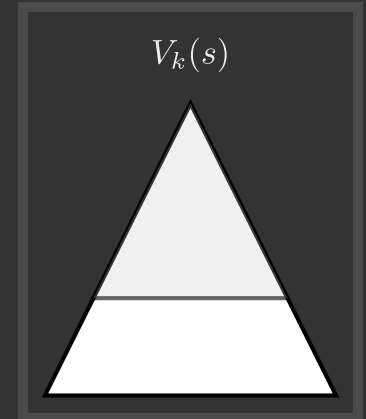
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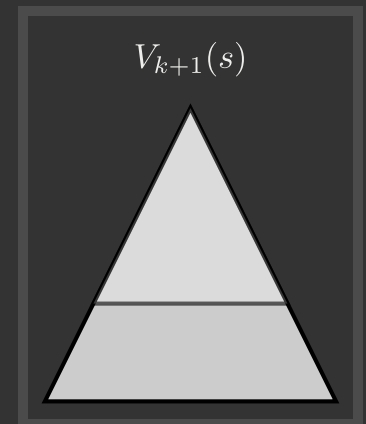
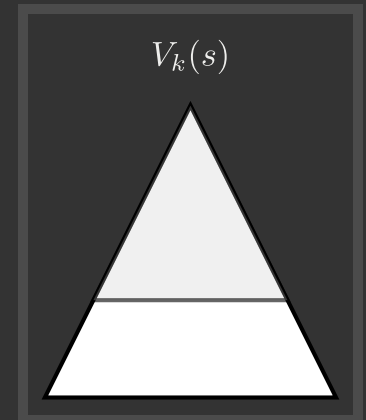
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Case 2: If the discount is less than 1

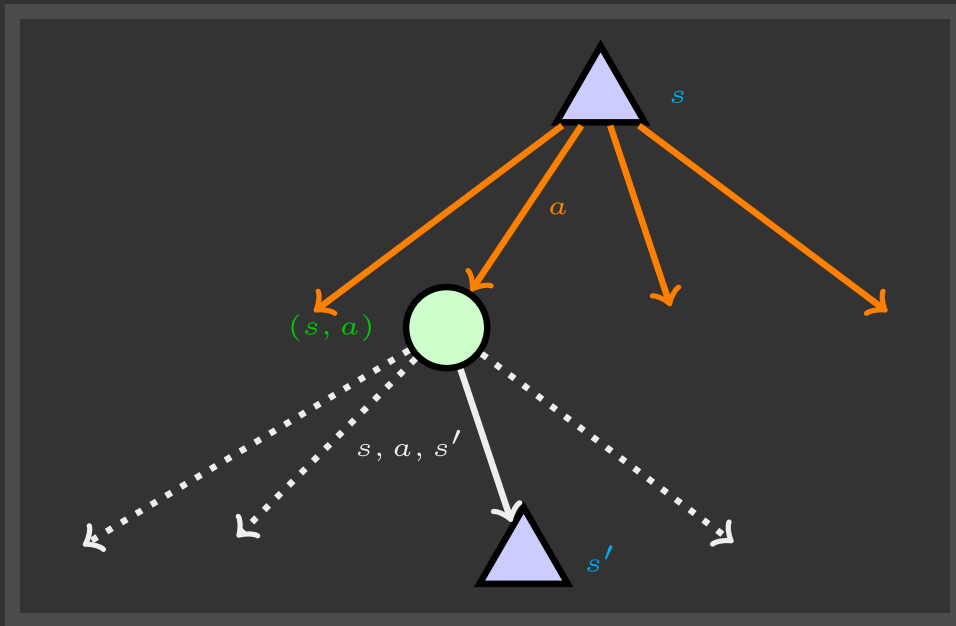
- Sketch: For any state V_k and V_{k+1} can be viewed as depth $k + 1$ expectimax results in nearly identical search trees
- The difference is that on the bottom layer, V_{k+1} has actual rewards while V_k has zeros
- That last layer is at best all R_{MAX}
- It is at worst R_{MIN}
- But everything is discounted by γ^k that far out
- So V_k and V_{k+1} are at most $\gamma^k \max |R|$ different
- So as k increases, the values converge



POLICY EVALUATION

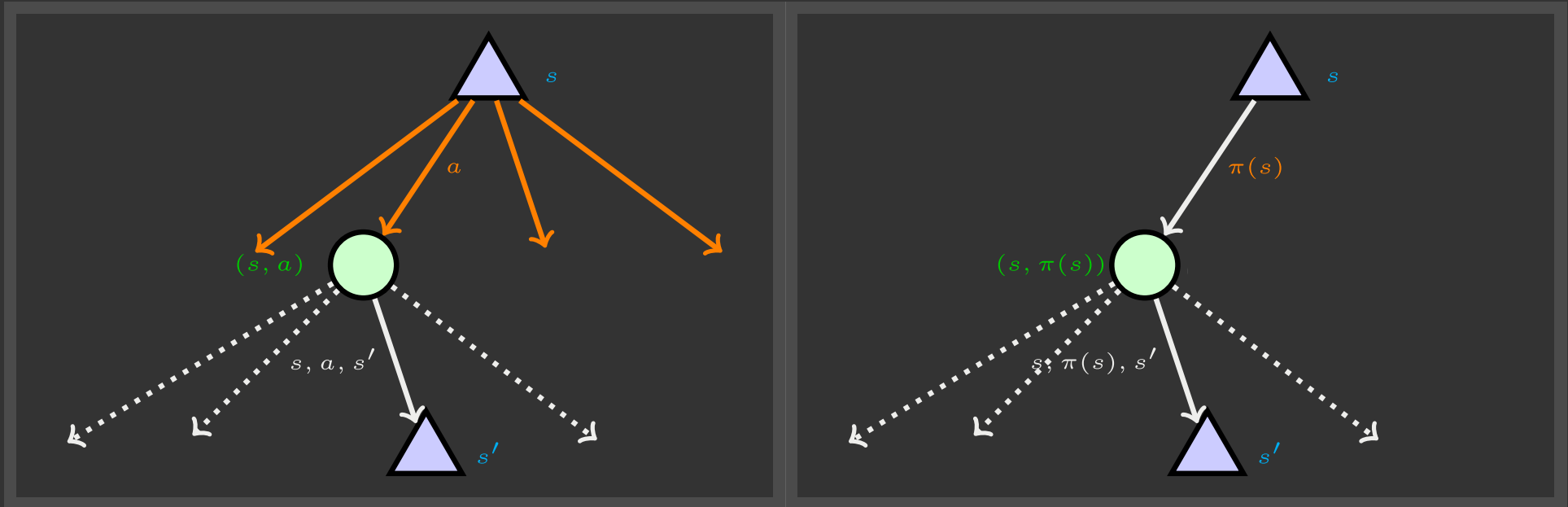


FIXED POLICIES



Expectimax trees max over all actions to compute the optimal values

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If we fixed some policy $\pi(s)$, then the tree would be simpler — only one action per state

- ... though the tree's value would depend on which policy we fixed

UTILITIES FOR A FIXED POLICY

Another basic operation: compute the utility of a state $\mathbf{s}$ under a fixed (generally non-optimal) policy

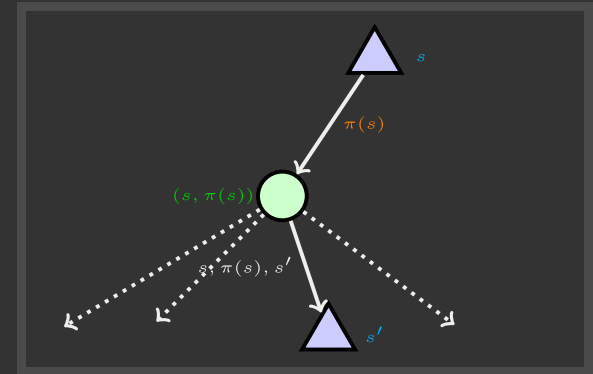


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$V^\pi(\mathbf{s}) =$ expected total discounted rewards starting in $\mathbf{s}$ and following π



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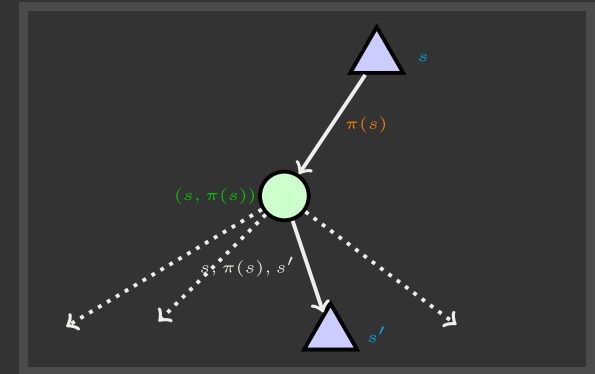
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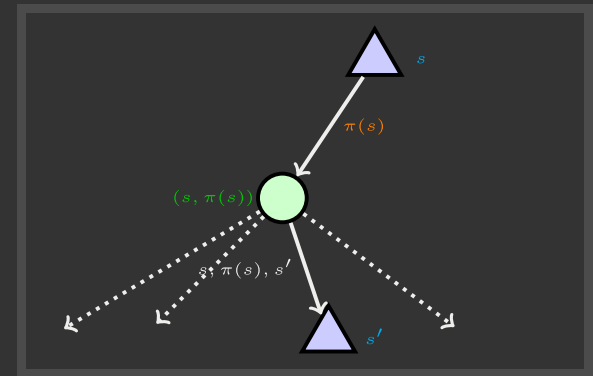
Recursive relation (one-step look-ahead / Bellman equation):

$$V^\pi(\mathbf{s}) = \sum_{\mathbf{s}'} T(\mathbf{s}, a, \mathbf{s}') [R(\mathbf{s}, a, \mathbf{s}') + \gamma V^\pi(\mathbf{s}')]$$



POLICY EVALUATION

How do we calculate the V 's for a fixed policy π ?

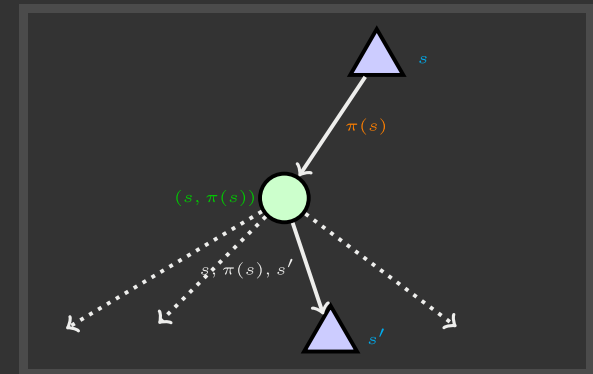


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Idea 1: Turn recursive Bellman equations into updates (like value iteration)

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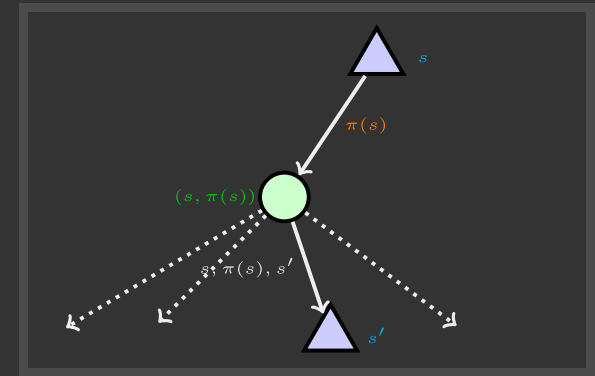
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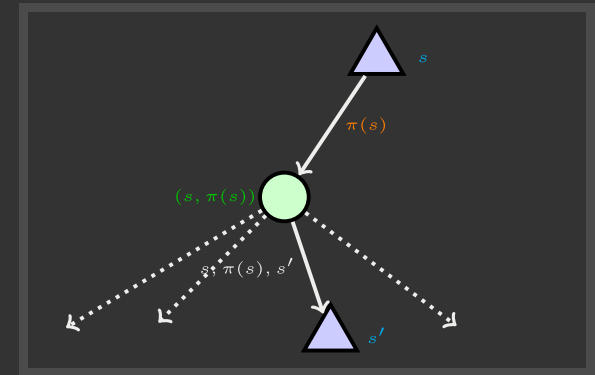
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Idea 2: Without the maxes, the Bellman equations are just a linear system

- Solve with Matlab (or your favorite linear system solver)



POLICY EXTRACTION



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This is called [policy extraction](#), since it gets the policy implied by the values.



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Important lesson: actions are easier to select from q-values than values !!



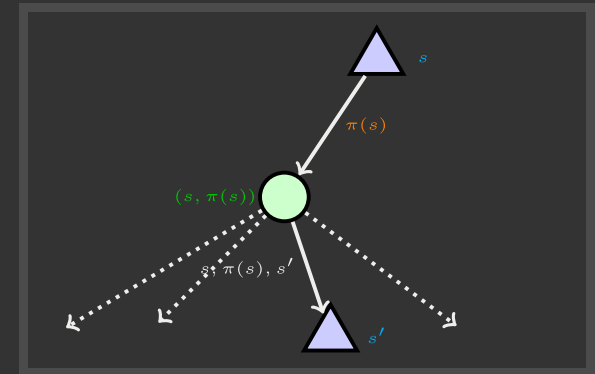
POLICY ITERATION



PROBLEMS WITH VALUE ITERATION

Value iteration repeats the Bellman updates:

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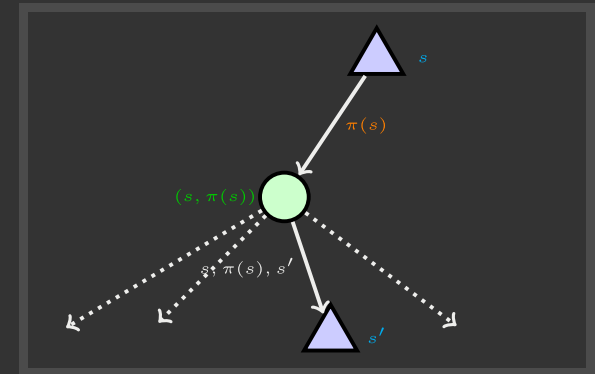
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Problem 1: It is slow — $\mathcal{O}(S^2A)$ per iteration

Problem 2: The "max" at each state rarely changes

Problem 3: The policy often converges long before the values



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Alternative approach for optimal values:



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- **Step 1: Policy evaluation:** calculate utilities for some fixed policy (not optimal utilities) until convergence
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- Repeat steps until policy converges



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This is policy iteration

- It is still optimal !!
- Can converge (much) faster under some conditions



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Evaluation: For fixed current policy π , find values with policy evaluation:



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- Iterate until values converge:

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Improvement: For fixed values, get a better policy using policy extraction



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Evaluation: For fixed current policy π , find values with policy evaluation:

- Iterate until values converge:

$$V_{k+1}^{\pi_i}(s) = \sum_{s'} T(s, \pi_i(s), s') [R(s, \pi_i(s), s') + \gamma V_k^{\pi_i}(s')]$$

Improvement: For fixed values, get a better policy using policy extraction

- One-step look-ahead:

$$\pi_{i+1}(s) = \operatorname{argmax}_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^{\pi_i}(s')]$$



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Both are dynamic programs for solving MDPs



SUMMARY: MDP ALGORITHMS

So you want to ...

- Compute optimal values: use value iteration or policy iteration
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SUMMARY: MDP ALGORITHMS

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These all look the same ...

- They basically are — they are all variations of Bellman updates
- They all use one-step lookahead expectimax fragments
- They differ only in whether we plug in a fixed policy or max over actions



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